

Document ID: TR-010

Revision: 1.0

Related Procedure: TP-010: Cybersecurity Verification Procedure

Verification Event: VE-005

Verification Method: Test

Date: 17 July 2026

Test Engineer: Serena Green

1. Purpose

Document the results of executing TP-010 to verify that the TOMTOM Ground Control Station satisfies the allocated cybersecurity requirements.

2. Referenced Documents

- TP-010 Cybersecurity Verification Procedure
- Verification Workbook
- GCS Requirements Specification
- Ground Control Station Interface Control Document
- System Requirements Specification

3. Configuration Baseline

Item	Configuration
Computer	ASUS G73S
Operating System	Linux Mint 22.3 Cinnamon (64- bit)
QGroundControl Version	V5.0.3-1109-g098330170 64-bit
Python Version	3.12.3
Git Revision	N/A
Procedure Revision	TP-010 Rev. 1.0
Vehicle Configuration	Not Connected (GCS Standalone Test)
Telemetry Radio	N/A- Not Installed
Date	17 July 2026

4. Test Personnel

Name	Role
Serena Green	Test Engineer
Alexander Green	Observer

5. Test Results

Verification Item	Expected	Actual	Pass/Fail
Authorized Login	Verify administrator authentication	Administrator authentication was verified successfully	Pass
Unauthorized Login	Invalid credentials are rejected and access to protected functions is denied.	Invalid credentials were rejected and access to protected functions was denied.	Pass
Admin Access	Administrative functions are accessible only after successful administrator authentication.	Administrative functions were accessible only after successful administrator authentication.	Pass
Encryption	Protected mission data is stored or transmitted using the configured operating system or application security controls, where applicable.	Upon startup, the system prompted for the LUKS disk encryption passphrase before loading Linux Mint. After entering the correct passphrase, the operating system booted successfully.	Pass
Event Logs	Security-relevant events (e.g., login attempts and configuration changes) are recorded with timestamps.	Opened terminal, executed <code>journalctl --since "10 minutes ago" grep "session opened"</code> , and verified <code>pam_unix(sudo:session)</code> entries were returned.	Pass
Mission Data	Unauthorized users cannot modify or delete protected mission data. Authorized users retain full access.	Mission planning functionality was unavailable in the current standalone GCS configuration. Mission data verification will be completed during integrated system testing with a connected flight controller.	N/A

Audit Log	Audit logs are successfully created, retained, and readable following test activities.	The Linux system journal was examined using journalctl following Ground Control Station testing. Timestamped log entries associated with QGroundControl execution and operation were successfully recorded and remained accessible for review. The audit log was readable, retained historical events after test activities, and provided chronological records of application activity. These results demonstrate that audit logging functions as intended for the GCS operating environment.	Pass
Configuration Control	Unauthorized users are prevented from modifying protected system configuration settings.	An attempt to modify a protected system configuration location (/etc) without administrative privileges was denied by the operating system. The system returned a permission error, preventing unauthorized modification of protected configuration files. Administrative access was required to perform configuration changes, demonstrating effective configuration control.	Pass
Media Access	Authorized storage media can be accessed normally; unauthorized or restricted access is denied according to system configuration.	Linux OS successfully detected and mounted the USB flash drive, allowing the authorized user to access removable storage normally. An attempt to modify a protected system storage location without	Pass

		administrative privileges was rejected with a "Permission denied" error, thus demonstrating that access controls are enforced according to the configured permission model.	
Shutdown	Normal system shutdown completes without data corruption, and security logs and mission data remain intact after restart.	Normal system shutdown completed without data corruption. Security logs and mission data remained intact after rebooting system.	Pass

6. Requirement Verification Summary

Requirement	Result	Evidence
SYS-SEC-001	PASS	Fig. A-2, Fig. A-3
SYS-SEC-002	PASS	Fig. A-6
SYS-SEC-004	PASS	Fig. A-5
SYS-CYB-001	PASS	Fig. A-8
SYS-CYB-002	PASS	Fig. A-2, Fig. A-3
SYS-CYB-003	PASS	Fig. A-6
SYS-CYB-006	PASS	Fig. A-4, Fig. A-5
SYS-DAT-004	PASS	Test Observation
SYS-MNT-002	PASS	Fig. A-6

7. Deviations

No deviations were experienced during testing.

8. Observations

System responsiveness was reduced during testing; however, this did not affect successful completion of the verification activities.

9. Discrepancies

No discrepancies between expected and actual outcomes were detected or experienced.

10. Overall Assessment

TP-010 was executed in accordance with Revision 1.0. All applicable cybersecurity requirements were successfully verified. Remaining verification activities requiring an integrated air vehicle were deferred to the system integration phase.

11. Evidence References

Figure	Description
Figure A-1	Ground Control Station test configuration.
Figure A-2	Administrator authentication prompt displayed prior to execution of privileged system commands.
Figure A-3	Successful administrator authentication permitting execution of privileged commands.
Figure A-4	Linux system journal displaying timestamped security-relevant events recorded during verification testing.
Figure A-5	Audit log entries associated with QGroundControl execution retained within the Linux system journal.
Figure A-6	Authorized removable storage device successfully detected and mounted by the operating system.
Figure A-7	Authorized removable media access verified and unauthorized modification of protected system storage denied by the operating system.
Figure A-8	Unauthorized attempt to modify a protected system configuration file denied by the operating system.

12. Approval

Role	Name	Signature	Date
Test Engineer	Serena Green	Serena Green	17 July 2026
Reviewer			

Appendix A: Verification Evidence

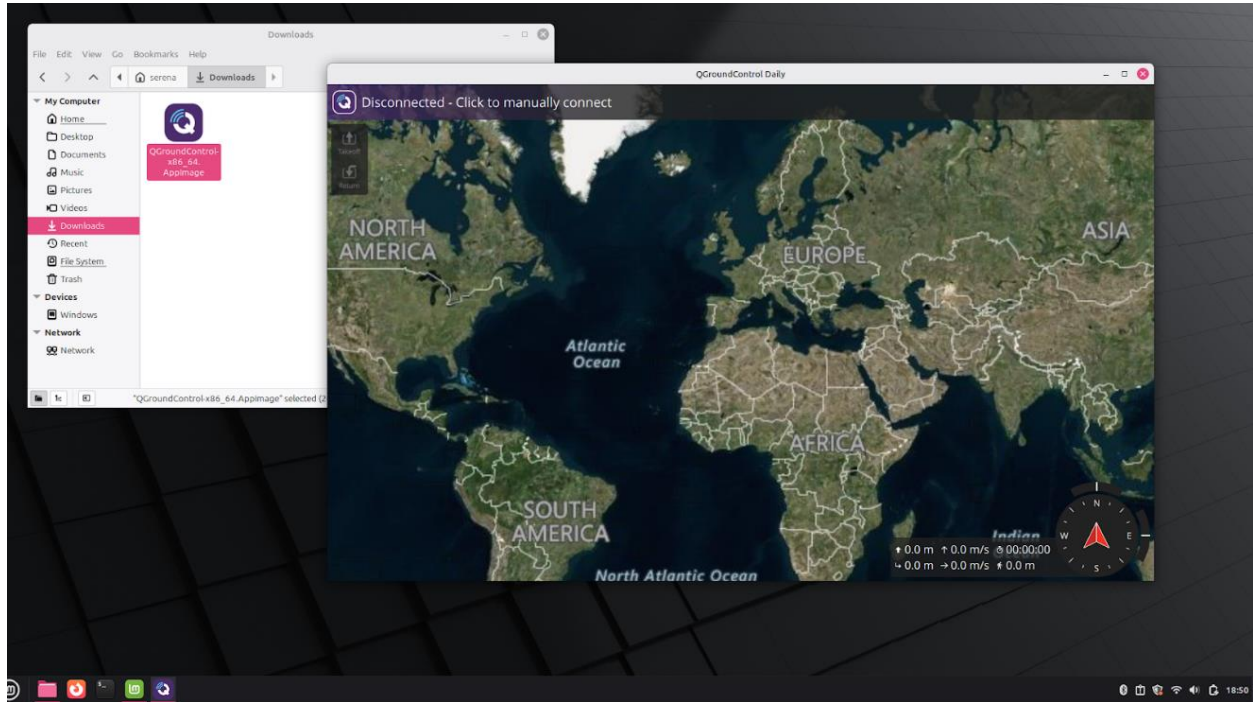


Figure 1-A. Ground Control Station configured for standalone cybersecurity verification using Linux Mint 22.3 and QGroundControl.

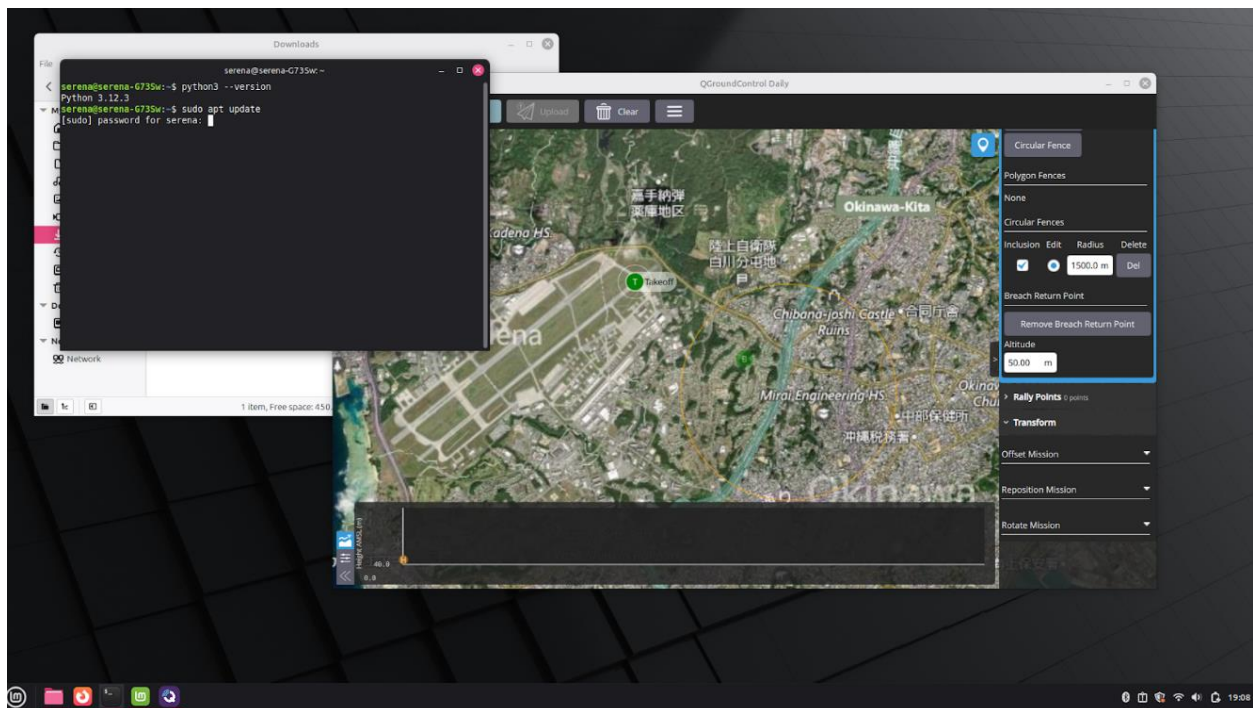


Figure 2-A. Administrator authentication was required before execution of privileged system

commands.

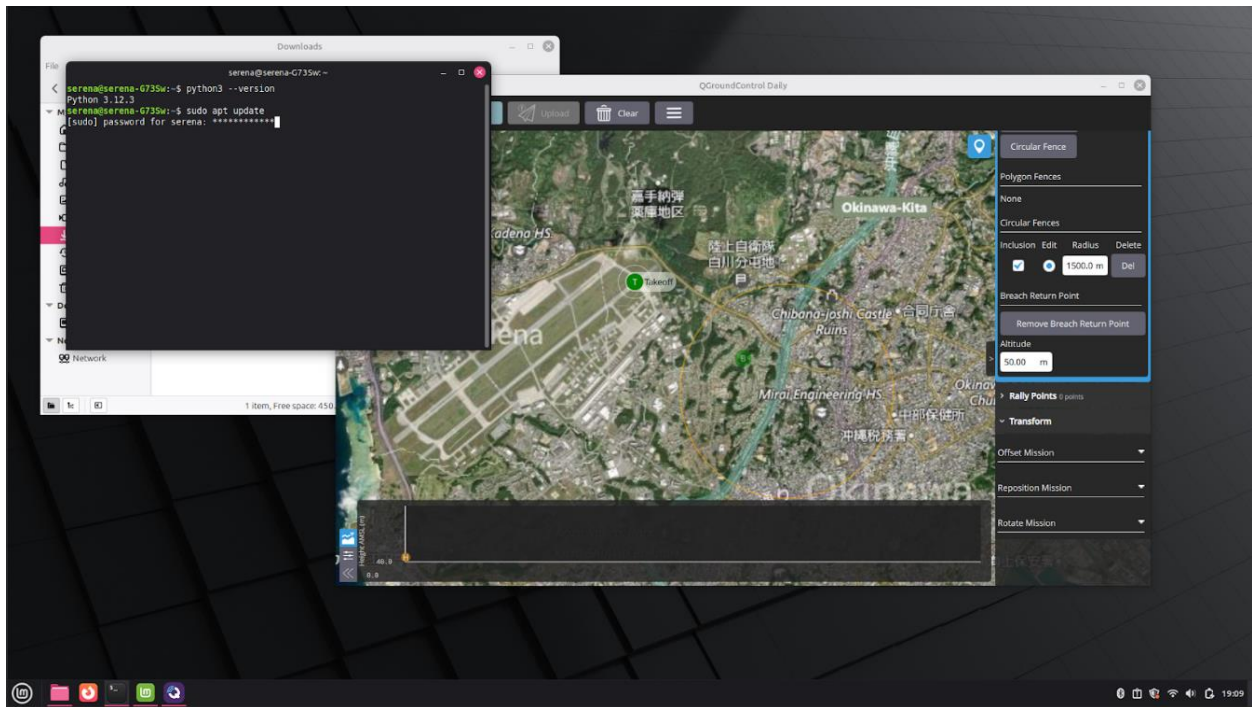


Figure 3-A. Administrator credentials accepted by the operating system, permitting execution of privileged commands. This verifies that protected administrative functions are accessible only following successful authentication.

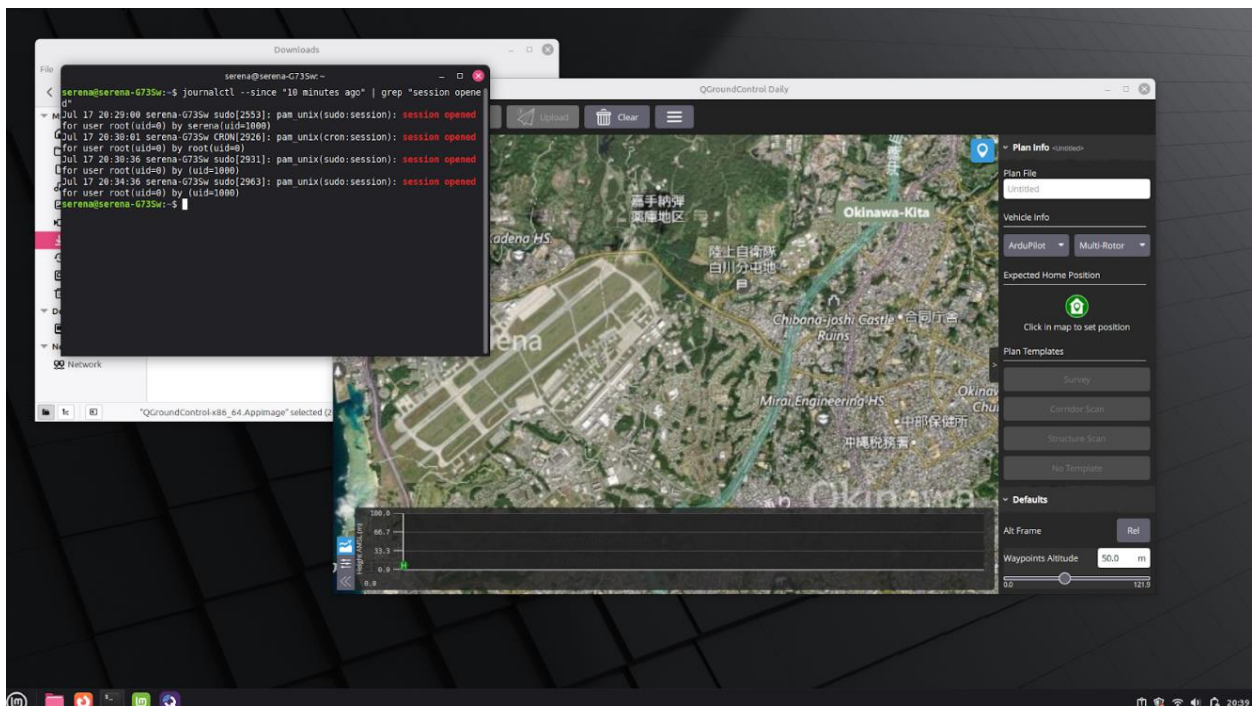


Figure 4-A. Linux system journal displaying timestamped security-related events generated

during testing. Session authentication records confirm that security-relevant events were successfully recorded by the operating system.

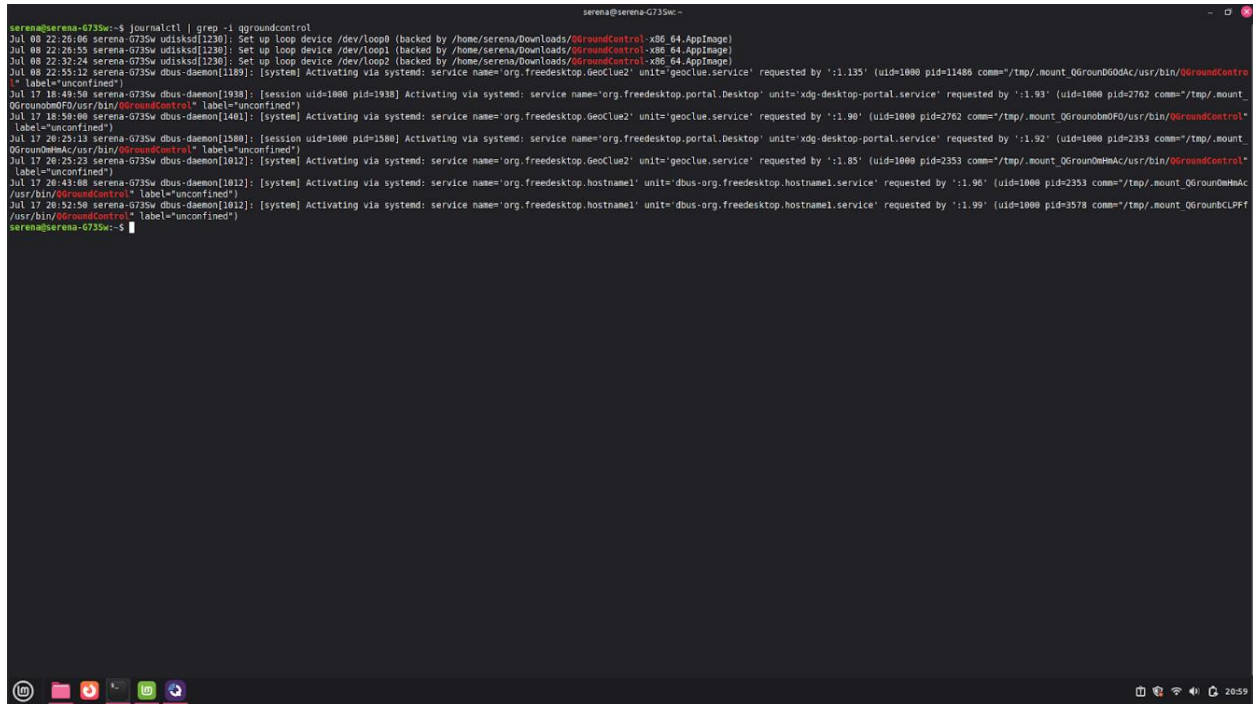
```
serena@serena-G735w:~$ journalctl | grep -i qgroundcontrol
Jul 08 22:26:06 serena-G735w udisksd[1230]: Set up loop device /dev/loop0 (backed by /home/serena/Downloads/QGroundControl-x86_64.AppImage)
Jul 08 22:26:55 serena-G735w udisksd[1230]: Set up loop device /dev/loop1 (backed by /home/serena/Downloads/QGroundControl-x86_64.AppImage)
Jul 08 22:32:24 serena-G735w udisksd[1230]: Set up loop device /dev/loop2 (backed by /home/serena/Downloads/QGroundControl-x86_64.AppImage)
Jul 08 22:55:12 serena-G735w dbus-daemon[1189]: [system] Activating via systemd: service name='org.freedesktop.GeoClue2' unit='geoclue.service' requested by '1.135' (uid=1000 pid=11486 comm="/tmp/.mount_QGroundControl0/usr/bin/QGroundControl" label="unconfined")
Jul 17 18:49:50 serena-G735w dbus-daemon[1938]: [session uid=1000 pid=1938] Activating via systemd: service name='org.freedesktop.portal.Desktop' unit='xdg-desktop-portal.service' requested by '1.93' (uid=1000 pid=2762 comm="/tmp/.mount_QGroundControl0/usr/bin/QGroundControl" label="unconfined")
Jul 17 18:50:00 serena-G735w dbus-daemon[1401]: [system] Activating via systemd: service name='org.freedesktop.GeoClue2' unit='geoclue.service' requested by '1.190' (uid=1000 pid=2762 comm="/tmp/.mount_QGroundControl0/usr/bin/QGroundControl" label="unconfined")
Jul 17 20:25:13 serena-G735w dbus-daemon[1580]: [session uid=1000 pid=1580] Activating via systemd: service name='org.freedesktop.portal.Desktop' unit='xdg-desktop-portal.service' requested by '1.92' (uid=1000 pid=2353 comm="/tmp/.mount_QGroundControl0/usr/bin/QGroundControl" label="unconfined")
Jul 17 20:25:23 serena-G735w dbus-daemon[1012]: [system] Activating via systemd: service name='org.freedesktop.GeoClue2' unit='geoclue.service' requested by '1.85' (uid=1000 pid=2353 comm="/tmp/.mount_QGroundControl0/usr/bin/QGroundControl" label="unconfined")
Jul 17 20:43:08 serena-G735w dbus-daemon[1012]: [system] Activating via systemd: service name='org.freedesktop.hostnamed' unit='dbus-org.freedesktop.hostnamed.service' requested by '1.96' (uid=1000 pid=2353 comm="/tmp/.mount_QGroundControl0/usr/bin/QGroundControl" label="unconfined")
Jul 17 20:52:50 serena-G735w dbus-daemon[1012]: [system] Activating via systemd: service name='org.freedesktop.hostnamed' unit='dbus-org.freedesktop.hostnamed.service' requested by '1.99' (uid=1000 pid=3578 comm="/tmp/.mount_QGroundControl0/usr/bin/QGroundControl" label="unconfined")
serena@serena-G735w:~$ touch /etc/test.config
touch: cannot touch '/etc/test.config': Permission denied
serena@serena-G735w:~$ lsblk
NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda           8:0 0 465.8G 0 disk 
├─sda1        8:1 0 512M 0 part /boot/efi
├─sda2        8:2 0 1.7G 0 part /boot
├─sda3        8:3 0 463.6G 0 part 
└─sda3 crypt 252:0 0 463.6G 0 crypt 
   └─vgmint-root 252:1 0 461.7G 0 lvm /
      └─vgmint-swap 252:2 0 1.9G 0 lvm [SWAP]
sdb           8:16 0 232.9G 0 disk 
├─sdb1        8:17 0 100M 0 part 
├─sdb2        8:18 0 10M 0 part 
├─sdb3        8:19 0 231.9G 0 part 
├─sdb4        8:20 0 900M 0 part 
├─sdc         8:32 1 28.9G 0 disk 
└─sdc1        8:33 1 28.9G 0 part /media/serena/OPENMEDIAVA
sr0           11:0 1 1024M 0 rom
```

Figure 5-A. Authorized removable storage device successfully detected and mounted by the operating system. The mounted device was accessible to the authorized user while protected system storage remained subject to operating system access controls.

```
serena@serena-G735w:~$ journalctl | grep -i qgroundcontrol
Jul 08 22:26:06 serena-G735w udisksd[1230]: Set up loop device /dev/loop0 (backed by /home/serena/Downloads/QGroundControl-x86_64.AppImage)
Jul 08 22:26:55 serena-G735w udisksd[1230]: Set up loop device /dev/loop1 (backed by /home/serena/Downloads/QGroundControl-x86_64.AppImage)
Jul 08 22:32:24 serena-G735w udisksd[1230]: Set up loop device /dev/loop2 (backed by /home/serena/Downloads/QGroundControl-x86_64.AppImage)
Jul 08 22:55:12 serena-G735w dbus-daemon[1189]: [system] Activating via systemd: service name='org.freedesktop.GeoClue2' unit='geoclue.service' requested by '1.135' (uid=1000 pid=11486 comm="/tmp/.mount_QGroundControl0/usr/bin/QGroundControl" label="unconfined")
Jul 17 18:49:50 serena-G735w dbus-daemon[1938]: [session uid=1000 pid=1938] Activating via systemd: service name='org.freedesktop.portal.Desktop' unit='xdg-desktop-portal.service' requested by '1.93' (uid=1000 pid=2762 comm="/tmp/.mount_QGroundControl0/usr/bin/QGroundControl" label="unconfined")
Jul 17 18:50:00 serena-G735w dbus-daemon[1401]: [system] Activating via systemd: service name='org.freedesktop.GeoClue2' unit='geoclue.service' requested by '1.190' (uid=1000 pid=2762 comm="/tmp/.mount_QGroundControl0/usr/bin/QGroundControl" label="unconfined")
Jul 17 20:25:13 serena-G735w dbus-daemon[1580]: [session uid=1000 pid=1580] Activating via systemd: service name='org.freedesktop.portal.Desktop' unit='xdg-desktop-portal.service' requested by '1.92' (uid=1000 pid=2353 comm="/tmp/.mount_QGroundControl0/usr/bin/QGroundControl" label="unconfined")
Jul 17 20:25:23 serena-G735w dbus-daemon[1012]: [system] Activating via systemd: service name='org.freedesktop.GeoClue2' unit='geoclue.service' requested by '1.85' (uid=1000 pid=2353 comm="/tmp/.mount_QGroundControl0/usr/bin/QGroundControl" label="unconfined")
Jul 17 20:43:08 serena-G735w dbus-daemon[1012]: [system] Activating via systemd: service name='org.freedesktop.hostnamed' unit='dbus-org.freedesktop.hostnamed.service' requested by '1.96' (uid=1000 pid=2353 comm="/tmp/.mount_QGroundControl0/usr/bin/QGroundControl" label="unconfined")
Jul 17 20:52:50 serena-G735w dbus-daemon[1012]: [system] Activating via systemd: service name='org.freedesktop.hostnamed' unit='dbus-org.freedesktop.hostnamed.service' requested by '1.99' (uid=1000 pid=3578 comm="/tmp/.mount_QGroundControl0/usr/bin/QGroundControl" label="unconfined")
serena@serena-G735w:~$ touch /etc/test.config
touch: cannot touch '/etc/test.config': Permission denied
serena@serena-G735w:~$ lsblk
NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda           8:0 0 465.8G 0 disk 
├─sda1        8:1 0 512M 0 part /boot/efi
├─sda2        8:2 0 1.7G 0 part /boot
├─sda3        8:3 0 463.6G 0 part 
└─sda3 crypt 252:0 0 463.6G 0 crypt 
   └─vgmint-root 252:1 0 461.7G 0 lvm /
      └─vgmint-swap 252:2 0 1.9G 0 lvm [SWAP]
sdb           8:16 0 232.9G 0 disk 
├─sdb1        8:17 0 100M 0 part 
├─sdb2        8:18 0 10M 0 part 
├─sdb3        8:19 0 231.9G 0 part 
├─sdb4        8:20 0 900M 0 part 
├─sdc         8:32 1 28.9G 0 disk 
└─sdc1        8:33 1 28.9G 0 part /media/serena/OPENMEDIAVA
sr0           11:0 1 1024M 0 rom

serena@serena-G735w:~$ touch /media/serena/OPENMEDIAVA/media_access_test.txt
serena@serena-G735w:~$ touch /root/media_access_test.txt
bash: touch /root/media_access_test.txt: No such file or directory
serena@serena-G735w:~$ echo "test" > /etc/media_access_test.txt
bash: /etc/media_access_test.txt: Permission denied
serena@serena-G735w:~$
```

Figure 6-A. *Authorized access to removable media was successfully demonstrated while attempts to modify protected system storage without administrative privileges resulted in a "Permission denied" response, verifying enforcement of operating-system access controls.*

A terminal window titled 'serena@serena-G735w: ~' displays the output of the command 'journalctl | grep -i qgroundcontrol'. The output shows several log entries from July 17 and 18, 2023, detailing the activation of various services by the 'org.freedesktop.portal.Desktop' unit. The services include 'xdg-desktop-portal.service', 'geoclue.service', and 'hostnamed.service'. Each entry specifies the user (uid=1000), process ID (pid), and the command being executed. The terminal window has a dark background and standard Linux window controls at the top. At the bottom, there is a taskbar with icons for various applications and system status indicators on the right, including a clock showing 20:59.

```
serena@serena-G735w: ~$ journalctl | grep -i qgroundcontrol
Jul 18 22:26:06 serena-G735w udisksd[1230]: Set up loop device /dev/loop0 (backed by /home/serena/Downloads/QGroundControl-x86_64.AppImage)
Jul 18 22:26:55 serena-G735w udisksd[1230]: Set up loop device /dev/loop1 (backed by /home/serena/Downloads/QGroundControl-x86_64.AppImage)
Jul 18 22:32:24 serena-G735w udisksd[1230]: Set up loop device /dev/loop2 (backed by /home/serena/Downloads/QGroundControl-x86_64.AppImage)
Jul 18 22:55:22 serena-G735w dbus-daemon[1189]: [system] Activating via systemd: service name='org.freedesktop.GeoClue2' unit='geoclue.service' requested by ':1.135' (uid=1000 pid=11486 comm="/tmp/.mount_QGroundControl0/usr/bin/QGroundControl" label="unconfined")
Jul 17 18:49:58 serena-G735w dbus-daemon[1938]: [session uid=1000 pid=1938] Activating via systemd: service name='org.freedesktop.portal.Desktop' unit='xdg-desktop-portal.service' requested by ':1.93' (uid=1000 pid=2762 comm="/tmp/.mount_QGroundControl0/usr/bin/QGroundControl" label="unconfined")
Jul 17 18:58:00 serena-G735w dbus-daemon[1401]: [system] Activating via systemd: service name='org.freedesktop.GeoClue2' unit='geoclue.service' requested by ':1.90' (uid=1000 pid=2762 comm="/tmp/.mount_QGroundControl0/usr/bin/QGroundControl" label="unconfined")
Jul 17 20:25:13 serena-G735w dbus-daemon[1580]: [session uid=1000 pid=1580] Activating via systemd: service name='org.freedesktop.portal.Desktop' unit='xdg-desktop-portal.service' requested by ':1.92' (uid=1000 pid=2353 comm="/tmp/.mount_QGroundControl0/usr/bin/QGroundControl" label="unconfined")
Jul 17 20:58:00 serena-G735w dbus-daemon[1812]: [system] Activating via systemd: service name='org.freedesktop.GeoClue2' unit='geoclue.service' requested by ':1.85' (uid=1000 pid=2353 comm="/tmp/.mount_QGroundControl0/usr/bin/QGroundControl" label="unconfined")
Jul 17 20:43:08 serena-G735w dbus-daemon[1812]: [system] Activating via systemd: service name='org.freedesktop.hostnamed' unit='dbus-org.freedesktop.hostnamed.service' requested by ':1.96' (uid=1000 pid=2353 comm="/tmp/.mount_QGroundControl0/usr/bin/QGroundControl" label="unconfined")
Jul 17 20:52:58 serena-G735w dbus-daemon[1812]: [system] Activating via systemd: service name='org.freedesktop.hostnamed' unit='dbus-org.freedesktop.hostnamed.service' requested by ':1.99' (uid=1000 pid=3578 comm="/tmp/.mount_QGroundControl0/usr/bin/QGroundControl" label="unconfined")
serena@serena-G735w: ~$
```

Figure 7-A. *System journal entries associated with QGroundControl execution were successfully recorded, retained, and remained readable following verification activities, demonstrating proper audit log functionality.*

```
serena@serena-G735w:~$ journalctl | grep -i qgroundcontrol
Jul 08 22:26:06 serena-G735w udisksd[1230]: Set up loop device /dev/loop0 (backed by /home/serena/Downloads/QGroundControl-x86_64.AppImage)
Jul 08 22:26:55 serena-G735w udisksd[1230]: Set up loop device /dev/loop1 (backed by /home/serena/Downloads/QGroundControl-x86_64.AppImage)
Jul 08 22:32:24 serena-G735w udisksd[1230]: Set up loop device /dev/loop2 (backed by /home/serena/Downloads/QGroundControl-x86_64.AppImage)
Jul 08 22:55:12 serena-G735w dbus-daemon[1189]: [system] Activating via systemd: service name='org.freedesktop.GeoClue2' unit='geoclue.service' requested by ':1.135' (uid=1000 pid=11486 comm="/tmp/.mount_QGroundControlAc/usr/bin/QGroundControl" label="unconfined")
Jul 17 18:49:58 serena-G735w dbus-daemon[1938]: [session uid=1000 pid=1938] Activating via systemd: service name='org.freedesktop.portal.Desktop' unit='xdg-desktop-portal.service' requested by ':1.93' (uid=1000 pid=2762 comm="/tmp/.mount_QGroundControlAc/usr/bin/QGroundControl" label="unconfined")
Jul 17 18:50:00 serena-G735w dbus-daemon[1401]: [system] Activating via systemd: service name='org.freedesktop.GeoClue2' unit='geoclue.service' requested by ':1.90' (uid=1000 pid=2762 comm="/tmp/.mount_QGroundControlAc/usr/bin/QGroundControl" label="unconfined")
Jul 17 20:25:13 serena-G735w dbus-daemon[1580]: [session uid=1000 pid=1580] Activating via systemd: service name='org.freedesktop.portal.Desktop' unit='xdg-desktop-portal.service' requested by ':1.92' (uid=1000 pid=2353 comm="/tmp/.mount_QGroundControlAc/usr/bin/QGroundControl" label="unconfined")
Jul 17 20:25:23 serena-G735w dbus-daemon[1812]: [system] Activating via systemd: service name='org.freedesktop.GeoClue2' unit='geoclue.service' requested by ':1.85' (uid=1000 pid=2353 comm="/tmp/.mount_QGroundControlAc/usr/bin/QGroundControl" label="unconfined")
Jul 17 20:43:08 serena-G735w dbus-daemon[1812]: [system] Activating via systemd: service name='org.freedesktop.hostname1' unit='dbus-org.freedesktop.hostname1.service' requested by ':1.96' (uid=1000 pid=2353 comm="/tmp/.mount_QGroundControlAc/usr/bin/QGroundControl" label="unconfined")
Jul 17 20:52:58 serena-G735w dbus-daemon[1812]: [system] Activating via systemd: service name='org.freedesktop.hostname1' unit='dbus-org.freedesktop.hostname1.service' requested by ':1.99' (uid=1000 pid=3578 comm="/tmp/.mount_QGroundControlAc/usr/bin/QGroundControl" label="unconfined")
serena@serena-G735w:~$ touch /etc/test_config
touch: cannot touch '/etc/test_config': Permission denied
serena@serena-G735w:~$
```

Figure 8-A. Attempt to modify a protected system configuration location without administrative privileges was denied by the operating system. The resulting permission error verifies that unauthorized users cannot modify protected configuration settings.